

100 Regex Ideas for Your Craigslist Cars

OPIM 5512 · Module 3 · idea sheet for Lab 3, A06 and your midterm

A **regular expression** (regex) is a tiny search pattern. Instead of searching for the exact text `2013`, you search for **"four digits on a line by themselves"**, and it finds the year in every ad at once. That is how your pipeline turns a messy web page into a row in a table.

Tonight (Lab 3) you only need ONE of these: the make/model fix (Way 2, idea 10.10). In **A06** you ship that one line to your pipeline through a pull request. The other 99 are a menu for later: new columns for your **midterm** price model (sections 1, 5, 6 and 8) and text clean-up before Gemini reads your ads in **A07** (section 9).

How to use this sheet - 10 sections × 10 ideas. Sections go from easiest to hardest, and so do the ideas inside each one (☆☆☆ easy · ★☆☆ medium · ★★★★★ stretch). - **Every pattern was tested** on Dr. Wanik's 40 Hartford ads (the Lab 3 backup data). The **Tested** line tells you how many ads it matched and what it grabbed. Your area will be different, and that is the point: test on YOUR cars. - A low hit count is not always a bug. Sometimes the info just is not in most ads (not many sellers write `obo`). - Try them on regex101.com (pick the **Python** flavor on the left, paste an ad from your `my_cars` folder), or run all 100 on your own cars in the [Try-It notebook](#). - Stuck on a pattern? Paste it into Claude and ask: *"Explain this regex piece by piece, like I'm new to regex."*

Start here: the regex alphabet

You type	It means	Example	Matches
<code>abc</code>	exactly these letters	<code>honda</code>	honda
<code>.</code>	any one character (except a line break)	<code>c.r</code>	car, cur, c9r
<code>\d</code>	one digit 0-9	<code>\d\d</code>	42
<code>\w</code>	one word character (letter, digit, <code>_</code>)	<code>\w\w\w</code>	fwd
<code>\s</code>	one whitespace (space, tab, line break)	<code>a\s b</code>	a b
<code>\D \W \S</code>	capital = the OPPOSITE (not digit, etc.)	<code>\D</code>	\$
<code>\n</code>	a line break	<code>odometer:\n</code>	the end of the label line
<code>[abc]</code>	one of these characters	<code>gr[ae]y</code>	gray, grey
<code>[a-z] [0-9]</code>	one character in a range	<code>[A-Z]</code>	any capital
<code>[^abc]</code>	one character NOT in the list	<code>[^,]</code>	anything but a comma
<code>+</code>	one or more of the thing before	<code>\d+</code>	7, 170000
<code>*</code>	zero or more	<code>\d*</code>	(nothing), 12
<code>?</code>	optional (zero or one)	<code>miles?</code>	mile, miles
<code>{4} {1,3}</code>	exactly 4 / between 1 and 3	<code>\d{4}</code>	2013
<code>A B</code>	A or B	<code>obo firm</code>	obo, firm
<code>()</code>	a capture group: the part you want back	<code>(\d+) miles</code>	group(1) = 170000
<code>(?:)</code>	a group that does NOT capture	<code>(?:19 20)\d\d</code>	2013
<code>^ \$</code>	start / end of a line (with <code>re.M</code>)	<code>^\d{4}\$</code>	a line that is only a year
<code>\A</code>	start of the whole text	<code>\A\d{4}</code>	the year in the title line
<code>\b</code>	word boundary (edge of a word)	<code>\bred\b</code>	red, not registered
<code>\\$ \. \(\</code>	backslash = "I mean the real symbol"	<code>\\$\d+</code>	\$4000

The 4 functions you need

Function	What it does	Gives you back
<code>re.search(p, text)</code>	finds the FIRST match	a match object <code>m</code> (or <code>None</code>): <code>m.group(0)</code> = whole match, <code>m.group(1)</code> = first group
<code>re.findall(p, text)</code>	finds EVERY match	a list of strings (or a list of tuples, if the pattern has 2+ groups)
<code>re.sub(p, new, text)</code>	finds and replaces	the new text
<code>re.split(p, text)</code>	cuts the text wherever <code>p</code> matches	a list of pieces

Flags go at the end: `re.search(p, text, re.I)`. Combine them with `|`, like `re.I | re.M`. - `re.I` = IGNORECASE (`honda` also matches `HONDA`) - `re.M` = MULTILINE (`^` and `$` work on every line, not just the start and end of the text) - `re.S` = DOTALL (`.` can also match a line break)

Always write patterns as raw strings: `r"..."`. The `r` stops Python from treating `\n` or `\d` as its own escape codes.

Try any idea on your own cars (Colab)

After Lab 3 Part 2b has made your `my_cars` folder, paste this into a new cell:

```
import glob
import re

pattern = r"odometer:\n([\d,]+)" # paste any idea from this sheet here
for path in sorted(glob.glob("/content/my_cars/*.txt")):
    with open(path, encoding="utf-8") as f:
        text = f.read()
        m = re.search(pattern, text)
        if m:
            print(path[-40:], "->", m.group(1))
        else:
            print(path[-40:], "-> no match")
```

The 10 sections

1. The details block (label: value)
2. Year
3. Dates, times and IDs
4. Price
5. Mileage
6. Paint color
7. Location
8. What the seller says (flags for your model)
9. Clean-up and privacy (`re.sub`)

10. Make and model (the boss level)

1. The details block (label: value)

The easiest wins. Craigslist prints a tidy block of `label:` lines, and the value is always on the NEXT line. Learn this pattern and you get half the dataset for free.

1.1 ★★★ Condition

```
re.search(r"condition:\n(.+)", text)
```

How it works: Find the words `condition:`, then `\n` (the line break), then `(.+)` = grab everything on the next line. The parentheses are a **capture group**: `.group(1)` gives you just what is inside them.

Tested: 37 of 40 Hartford ads → `good`, `like new`, `excellent`

1.2 ★★★ Title status

```
re.search(r"title status:\n(\w+)", text)
```

How it works: Same recipe, but `\w+` = one or more **word characters** (letters, digits, underscore). It stops at the end of the word, so you never pick up stray spaces.

Tested: 40 of 40 Hartford ads → `clean`, `rebuilt`, `missing`

1.3 ★★★ Transmission

```
re.search(r"transmission:\n(\w+)", text)
```

How it works: Same recipe again. Once you see the pattern `label:\n(\w+)`, you can copy it for any field.

Tested: 40 of 40 Hartford ads → `automatic`, `manual`

1.4 ★★★ Drive (fwd / rwd / 4wd)

```
re.search(r"drive:\n(\w+)", text)
```

How it works: `\w` includes digits, so `4wd` comes through whole.

Watch out: Case matters by default. `drive:` does not match `DRIVE:` unless you add `re.I`.

Tested: 39 of 40 Hartford ads → `fwd`, `rwd`, `4wd`

1.5 ★★★ Cylinders as a number

```
re.search(r"cylinders:\n(\d+)", text)
```

How it works: `\d+` = one or more **digits**. The value line says `4 cylinders`; `\d+` grabs the `4` and stops at the space. Wrap it in `int(...)` and it is ready for a model.

Tested: 40 of 40 Hartford ads → `4`, `8`, `6`

1.6 ★★★ Body type

```
re.search(r"^type:\n(\w+)", text, re.M)
```

How it works: `^` = **start of a line** (needs `re.M`, the MULTILINE flag).

Watch out: Without `^`, `type:` would also match the end of `prototype:` or `body type:`. Anchors keep you honest.

Tested: 36 of 40 Hartford ads → `SUV`, `truck`, `coupe`

1.7 ★★★ Paint color (only if listed)

```
re.search(r"paint color:\n(\w+)", text)
```

How it works: Same recipe, but this field is **optional**: many sellers skip it. When there is no match, `re.search` returns `None`, so check `if m:` before calling `.group(1)`.

Watch out: A low hit count is not a broken regex. It is missing data. Know the difference.

Tested: 34 of 40 Hartford ads → `red`, `blue`, `black`

1.8 ★★★ Only the NON-clean titles

```
re.search(r"title status:\n(?!clean)(\w+)", text)
```

How it works: `(?!clean)` is a **negative lookahead**: "only continue if the next word is NOT `clean`." It finds the rebuilt, salvage and missing titles, which are exactly the cars a buyer should worry about.

Tested: 2 of 40 Hartford ads → `rebuilt`, `missing`

1.9 ★★★ Every label and value at once

```
re.findall(r"^([a-z ]+):\n(.*?)$", text, re.M)
```

How it works: `[a-z ]+` = a **character class**: any lowercase letter or a space, one or more times. With two groups, `re.findall` returns a list of `(label, value)` pairs, and `dict(...)` turns that into a lookup table in one line.

Watch out: It also catches `posted:` (a date). Lowercase-only keeps out `Contact Information:`, which starts with a capital.

Tested: 40 of 40 Hartford ads → `[('condition', 'good'), ('cylinders', '4 cylind...)]`

1.10 ★★★ The details block, cut out whole

```
re.search(r"^(?:19|20)\d{2}\n.*\n((?:[a-z ]+:\n.+)+)", text, re.M)
```

How it works: `(?:...)` is a group that does **not** capture (it just groups). `(?:[a-z ]+:\n.+)+` = one label line plus one value line, repeated one or more times. The outer `( )` captures the whole block.

Watch out: Idea 1.9 is easier to use; this one shows how groups can repeat.

Tested: 36 of 40 Hartford ads → `condition:␣good␣cylinders:␣4 cylinders␣drive:...`,
`condition:␣good␣cylinders:␣8 cylinders␣drive:...`, `condition:␣like new␣cylinders:␣4 cylinders␣dr...`

2. Year

Every ad has the year in 2 or 3 places. The trick is grabbing the RIGHT 4 digits.

2.1 ★★★ Any 4 digits

```
re.search(r"\d{4}", text)
```

How it works: `\d` = a digit; `{4}` = exactly four of them. `re.search` returns the **first** match, which here is the year at the start of the title.

Watch out: Lucky! Try `re.findall` on the same pattern: you get the posted date, the rpm, the post id... `\d{4}` is not the same as a year.

Tested: 40 of 40 Hartford ads → `2020`, `1966`, `2004`

2.2 ★★★ Starts with 19 or 20

```
re.search(r"(19|20)\d{2}", text)
```

How it works: `(19|20)` = **alternation**: 19 OR 20. Then two more digits. `5600` rpm no longer counts.

Watch out: Use `m.group(0)` (the whole match). `m.group(1)` is only what is inside the parentheses, so `'20'`.

Tested: 40 of 40 Hartford ads → `2020`, `1966`, `2004`

2.3 ★★★ Whole numbers only

```
re.search(r"\b(19|20)\d{2}\b", text)
```

How it works: `\b` = a **word boundary** (the edge between a word and a non-word). It stops `2020` from matching inside `7974020201`.

Tested: 40 of 40 Hartford ads → `2020`, `1966`, `2004`

2.4 ★★★ Year at the very start of the file

```
re.search(r"^\A\d{4}", text)
```

How it works: `\A` = the **start of the whole text**. Line 1 of every file is the page title, and the title starts with the year.

Tested: 40 of 40 Hartford ads → `2020`, `1966`, `2004`

2.5 ★★★ A line that is ONLY a year

```
re.search(r"^((?:19|20)\d{2})$", text, re.M)
```

How it works: With `re.M`, `^` and `$` mean the start and end of **each line**. Craigslist puts the year alone on its own line just above the details block.

Tested: 40 of 40 Hartford ads → 2020, 1966, 2004

2.6 ★★★ Year from the title line

```
re.search(r"^(\\d{4}) .+ for sale by owner", text)
```

How it works: `.+` = any characters, one or more. Anchoring on the boilerplate `for sale by owner` means you only look at the title.

Tested: 40 of 40 Hartford ads → 2020, 1966, 2004

2.7 ★★★ The POSTING year (not the car year!)

```
re.search(r"Posted\\n(\\d{4})-", text)
```

How it works: Context decides meaning. `Posted` + line break + 4 digits + a dash gives the year of the ad itself.

Watch out: Use it for car age: 2026 minus the car year.

Tested: 40 of 40 Hartford ads → 2026

2.8 ★★★ Only plausible car years

```
re.search(r"\\b(19[5-9]|20[0-2]\\d)\\b", text)
```

How it works: `[5-9]` is a **range** inside a character class. This allows 1950-1999 or 2000-2029, and nothing else.

Tested: 40 of 40 Hartford ads → 2020, 1966, 2004

2.9 ★★★ Year between line breaks, with lookarounds

```
re.search(r"(?<=\\n)(?:19|20)\\d{2}(?>=\\n)", text)
```

How it works: `(?<=\\n)` = **lookbehind** (a line break must come before). `(?>=\\n)` = **lookahead** (a line break must come after). Neither is included in the match. It is the same result as 2.5, without the MULTILINE flag.

Tested: 40 of 40 Hartford ads → 2020, 1966, 2004

2.10 ★★★ Do the years disagree?

```
re.findall(r"\\b(?:19|20)\\d{2}\\b", text)
```

How it works: `re.findall` returns every year-looking number in the ad. Put them in a `set()`; if the title year and the description year differ, flag the ad. One Hartford seller lists a 1997 Wrangler whose description says 2008!

Watch out: Use `(?:19|20)`, not `(19|20)`. With a capture group, `findall` returns only the group, so you would get `'20'` instead of `'2008'`.

Tested: 40 of 40 Hartford ads → ['2020', '2026', '2020']

3. Dates, times and IDs

The easy structured stuff: always the same shape, so it is great practice with exact counts.

3.1 ★★★ The post id

```
re.search(r"post id: (\d+)", text)
```

How it works: Literal text `post id:` followed by digits. This is the ad's unique key; use it to remove duplicates.

Tested: 40 of 40 Hartford ads → `7974058269`, `7974064302`, `7974037488`

3.2 ★★★ A date

```
re.search(r"\d{4}-\d{2}-\d{2}", text)
```

How it works: 4 digits, dash, 2 digits, dash, 2 digits. The dash is just a normal character here.

Tested: 40 of 40 Hartford ads → `2026-09-27`, `2026-09-28`, `2026-09-26`

3.3 ★★★ Year, month and day as separate groups

```
re.search(r"(\d{4})-(\d{2})-(\d{2})", text)
```

How it works: Three capture groups: `.group(1)`, `.group(2)` and `.group(3)`, or all at once with `.groups()`.

Tested: 40 of 40 Hartford ads → `('2026', '09', '27')`, `('2026', '09', '28')`, `('2026', '09', '26')`

3.4 ★★★ A clock time

```
re.search(r"\b\d{2}:\d{2}\b", text)
```

How it works: Two digits, colon, two digits. The `\b` stops it from matching the middle of a longer number.

Tested: 40 of 40 Hartford ads → `21:01`, `21:21`, `20:06`

3.5 ★★★ Date AND time after 'Posted'

```
re.search(r"Posted\n(\d{4}-\d{2}-\d{2} \d{2}:\d{2})", text)
```

How it works: Glue the two patterns together with a space. `pd.to_datetime(...)` turns the result into a real timestamp.

Tested: 40 of 40 Hartford ads → `2026-09-27 21:01`, `2026-09-27 21:21`, `2026-09-27 20:06`

3.6 ★★★ Named groups

```
re.search(r"(?P<year>\d{4})-(?P<month>\d{2})-(?P<day>\d{2})", text)
```

How it works: `(?P<name>...)` gives a group a name. Then `m.group('month')` or `m.groupdict()` gives you a ready-made dict.

Tested: 40 of 40 Hartford ads → `('2026', '09', '27')`, `('2026', '09', '28')`, `('2026', '09', '26')`

3.7 ★★★ Hour of posting (a feature!)

```
re.search(r"Posted\n\d{4}-\d{2}-\d{2} (\d{2}):", text)
```

How it works: Capture only the hour. Do evening ads sell for more? Now you can check.

Tested: 40 of 40 Hartford ads → `21`, `20`, `22`

3.8 ★★★ How many photos

```
re.search(r"image 1 of (\d+)", text)
```

How it works: The gallery says `image 1 of 24`. More photos might mean a more serious seller, so this is a free feature for your price model.

Watch out: Ads with a single photo do not show this line, so no match means 1 photo.

Tested: 38 of 40 Hartford ads → `10`, `24`, `6`

3.9 ★★★ A VIN, anywhere

```
re.search(r"\b[A-HJ-NPR-Z0-9]{17}\b", text)
```

How it works: A VIN is exactly 17 characters and never uses I, O or Q. The character class `[A-HJ-NPR-Z0-9]` means A-H, J-N, P, R-Z or a digit, which is exactly the VIN alphabet, and `{17}` sets the length.

Tested: 4 of 40 Hartford ads (matches hidden: personal or identifying info)

3.10 ★★★ Run id and post id from a file path

```
re.search(r"scrapes/(\d{14})/([A-Za-z0-9]+)\.txt", text)
```

How it works: `\.` is an **escaped** dot (a real period; a bare `.` means any character). The run id is a 14-digit timestamp (YYYYMMDDHHMMSS), and the post id is the file name. Your scraper pulls the same id out of the ad link with `/view/d/[^/]+/([A-Za-z0-9]+)`, where `[^/]+` = anything except a slash.

Watch out: Tested on the `source_txt` column of `listings_master.csv`, not on the ad text.

Tested: 30 of 30 Hartford file paths → `('20260928155246', '24WkQVAv4khn6QWdTBQHF1')`, `('20260928155246', '4ysVPsYd2KTDr6rbicgnoV')`, `('20260928155246', '7dEEfyBPBV7RpUamCJ258v')`

4. Price

The dollar sign is a special character, commas break numbers apart, and sellers write prices every way imaginable.

4.1 ★★★ A dollar sign and digits

```
re.search(r"\$\d+", text)
```

How it works: `$` normally means "end of line", so to find a real dollar sign you **escape** it with a backslash: `\$`.

Watch out: On `$11,500` this returns just `$11`, because a comma is not a digit. See the next one.

Tested: 40 of 40 Hartford ads → `$11`, `$14`, `$7`

4.2 ★★★ Let commas in

```
re.search(r"\$[\d,]+", text)
```

How it works: `[\d,]` = a digit OR a comma. Now `$11,500` comes through whole.

Tested: 40 of 40 Hartford ads → `$11,500`, `$14,000`, `$7,600`

4.3 ★★★ Negotiable?

```
re.search(r"\b(obo|or best offer|best offer)\b", text, re.I)
```

How it works: Alternation plus `re.I` (IGNORECASE) catches `OB0`, `obo` and `or Best Offer`. That gives a yes/no feature.

Watch out: Order matters inside `( | )`: the longer option `or best offer` comes first, so it wins.

Tested: 5 of 40 Hartford ads → `OB0`, `or Best Offer`, `obo`

4.4 ★★★ Firm price?

```
re.search(r"\b(firm|non[- ]?negotiable)\b", text, re.I)
```

How it works: `[- ]?` = an optional dash or space, so it matches `non-negotiable`, `non negotiable` and `nonnegotiable`.

Tested: 8 of 40 Hartford ads → `FIRM`, `firm`

4.5 ★★★ Real thousands formatting

```
re.search(r"\$\d{1,3}(?:,\d{3})*", text)
```

How it works: 1-3 digits, then zero or more groups of `,` plus 3 digits (`*` = zero or more). It won't accept junk like `$1,2,3`.

Tested: 40 of 40 Hartford ads → `$11,500`, `$14,000`, `$7,600`

4.6 ★★★ The price line in the header

```
re.search(r"^\$([\d,]+)$", text, re.M)
```

How it works: The official price sits alone on its own line. `^` and `$` (with `re.M`) demand that the whole line is the price.

Watch out: Here `$` at the end is the anchor, and `\$` at the start is the dollar sign. Same symbol, two meanings.

Tested: 40 of 40 Hartford ads → `11,500`, `14,000`, `7,600`

4.7 ★★★ The price after the title dash

```
re.search(r"-\\n\\$([\\d,]+)", text)
```

How it works: In the header, the title is followed by a line that is just `-`, then the price. Using the neighbor as a landmark is a common scraping trick.

Tested: 40 of 40 Hartford ads → `11,500`, `14,000`, `7,600`

4.8 ★★★ Clean it into a number

```
re.sub(r"D", "", "$11,500")
```

How it works: `\D` (capital D) = anything that is **not** a digit. Replace all of them with nothing: `re.sub(r"D", "", "$11,500")` gives `11500`. Then use `int(...)`.

Watch out: Careful with `$3.000` (European thousands) and `$11,500.99`. Look at your data before you trust a cleaner.

Result: `11500`

4.9 ★★★ Every dollar amount in the ad

```
re.findall(r"\$\s*:? \s*(\d[\d,.]*)", text)
```

How it works: `\s*` = any amount of whitespace (including none), and `:?` = an optional colon. That survives messy text like `$:3.000`. `findall` shows when the description price disagrees with the header price.

Tested: 40 of 40 Hartford ads → `['11,500']`

4.10 ★★★ '14k OBO' style prices

```
re.search(r"\b(\d+(?:\.\d)?)\s?k\s*(?:obo|or best offer|firm)", text, re.I)
```

How it works: A number, maybe a decimal (`(?:\.\d)?`), optional space, `k`, then a price word. Multiply by 1000.

Watch out: Without the price word you would also grab `130k` MILES. Context is everything.

Tested: 1 of 40 Hartford ads → `14`

5. Mileage

The odometer field is clean, but half the useful mileage info lives in the free text: `130k`, `88K miles`, `41,534 ORIGINAL MILES`.

5.1 ★★★ The odometer field

```
re.search(r"odometer:\n(.+)", text)
```

How it works: The details-block recipe from section 1. Start here.

Tested: 40 of 40 Hartford ads → 170,000 , 84,000 , 21,000

5.2 ★★★ Just the number

```
re.search(r"odometer:\n([\d,]+)", text)
```

How it works: `[\d,]+` keeps digits and commas and stops at anything else.

Tested: 40 of 40 Hartford ads → 170,000 , 84,000 , 21,000

5.3 ★★★ '... miles' in the text

```
re.search(r"([\d,]+) miles", text, re.I)
```

How it works: A number, a space, then `miles`, ignoring case, so `MILES` counts too.

Tested: 10 of 40 Hartford ads → 170,000 , 21,000 , 122,331

5.4 ★★★ The 'k' shorthand

```
re.search(r"\b(\d{2,3})\s?k\b", text, re.I)
```

How it works: 2-3 digits, optional space (`\s?`), then `k` as a whole word. Matches `130k`, `88K` and `180 k`.

Watch out: It also catches `14k 0B0` (a price!) and `Oil change every 10-13 k miles` (not the odometer). See 5.10.

Tested: 13 of 40 Hartford ads → 13 , 14 , 88

5.5 ★★★ 'k miles'

```
re.search(r"\b(\d{2,3})\s?k\s+miles\b", text, re.I)
```

How it works: Requiring `miles` after the `k` removes the price confusion, but misses ads that write just `130k`.

Tested: 7 of 40 Hartford ads → 13 , 88 , 128

5.6 ★★★ 'ORIGINAL miles', even misspelled

```
re.search(r"([\d,]+)\s+or?i?ginal\s+miles", text, re.I)
```

How it works: `?` = the previous character is **optional**. `or?i?ginal` matches `original` and the seller's `ORIGINAL`. Typos are part of real data.

Tested: 4 of 40 Hartford ads → 98,000 , 189,000 , 19,925

5.7 ★★★ Bragging words

```
re.search(r"\b(?:only|just)\s+([\d,]+(?:\s?k)?)\b", text, re.I)
```

How it works: `only 21,000` and `ONLY 88K` are low-mileage brags. `(?:...)` groups `only|just` without capturing it.

Tested: 6 of 40 Hartford ads → `122,331`, `98,000`, `88K`

5.8 ★★★ Either format with alternation

```
re.search(r"\b(\d{1,3}(?:,\d{3})+|\d{2,3}\s?k)\s*miles", text, re.I)
```

How it works: `( A | B )`: A = comma-formatted (`170,000`), B = k-format (`88K`). Both must be followed by `miles`.

Tested: 15 of 40 Hartford ads → `13 k`, `21,000`, `122,331`

5.9 ★★★ Skip 'miles on the motor'

```
re.findall(r"([\d,]+)\s+miles(?:\s+on\s+(?:the\s+)?(?:motor|engine))", text, re.I)
```

How it works: `(?! ...)` = negative lookahead. `less than 1000 miles on motor` is about a new engine, not the car, so the lookahead throws it out.

Tested: 9 of 40 Hartford ads → `['170,000']`

5.10 ★★★ k-miles that are NOT prices

```
re.search(r"\b(\d{2,3})\s?k\b(?:!\s*(?:obo|or best|firm))", text, re.I)
```

How it works: The fix for 5.4: a `k` number that is not followed by a price word. Then `int(m.group(1)) * 1000` gives miles.

Tested: 12 of 40 Hartford ads → `13`, `88`, `128`

6. Paint color

Only some sellers fill in `paint color:`, so the rest of the color info hides in sentences like `BURGUNDY WITH TAN TWEED INTERIOUR`.

6.1 ★★★ The naive search

```
re.search(r"red", text, re.I)
```

How it works: Plain text is a valid regex. It finds `red` ...

Watch out: ...and also finds `red` inside `regist-ERED`, `cove-RED` and `tRED`. That is why we need 6.2.

Tested: 16 of 40 Hartford ads → `red`

6.2 ★★★ Whole word only

```
re.search(r"\bred\b", text, re.I)
```

How it works: `\b` on both sides = `red` must be a whole word. The most important fix on this sheet.

Tested: 11 of 40 Hartford ads → `red`

6.3 ★★★ The paint color field

```
re.search(r"paint color:\n(\w+)", text)
```

How it works: The details-block recipe. The most reliable, when it is there.

Tested: 34 of 40 Hartford ads → `red`, `blue`, `black`

6.4 ★★★ gray or grey

```
re.search(r"\bgr[ae]y\b", text, re.I)
```

How it works: `[ae]` = exactly one character, `a` OR `e`. One pattern, both spellings.

Tested: 11 of 40 Hartford ads → `Gray`, `GRAY`, `grey`

6.5 ★★★ A list of colors

```
re.search(r"\b(red|blue|black|white|silver|gr[ae]y|green|burgundy|tan|gold)\b", text, re.I)
```

How it works: Alternation inside a group, with word boundaries on the outside so every option is a whole word.

Tested: 33 of 40 Hartford ads → `Gray`, `red`, `blue`

6.6 ★★★ '___ exterior'

```
re.search(r"(\w+) exterior", text, re.I)
```

How it works: The word right before `exterior`. It finds `Gray exterior` and even `mica exterior` (from `claret mica`).

Tested: 2 of 40 Hartford ads → `Gray`, `mica`

6.7 ★★★ Shade words

```
re.search(r"\b(dark|light|bright|metallic)\s+(red|blue|gr[ae]y|green|silver)\b", text, re.I)
```

How it works: Two groups: the shade and the color. `Bright Red` becomes `('Bright', 'Red')`.

Tested: 2 of 40 Hartford ads → `('Bright', 'Red')`, `('DARK', 'GRAY')`

6.8 ★★★ Exterior/Interior pairs

```
re.search(r"\b(\w+)\s*/\s*(\w+)\s+(?:\w+\s+)?interiou?r\b", text, re.I)
```

How it works: BLUE/TAN LEATHER INTERIOUR gives ('BLUE', 'TAN'). \s*/\s* allows spaces around the slash, (?:\w+\s+)? allows one optional word (leather, cloth, velour), and interiou?r forgives the extra u.

Watch out: On Gray exterior / Black cloth interior it returns ('exterior', 'Black'). Close, but regex has no idea what words mean.

Tested: 5 of 40 Hartford ads → ('exterior', 'Black'), ('BLUE', 'TAN'), ('BURGUNDY', 'GRAY')

6.9 ★★★ 'COLOR WITH ... INTERIOR' lines

```
re.search(r"^( [a-z ]+?) with .*interiou?r", text, re.I | re.M)
```

How it works: +? is a **lazy** (non-greedy) repeat: it grabs as LITTLE as possible, so the group stops at the first with. DARK GRAY WITH CLOTH INTERIOUR gives DARK GRAY.

Tested: 3 of 40 Hartford ads → BURGUNDY, SOLID BODY, DARK GRAY

6.10 ★★★ Field first, text as backup

```
re.search(r"paint color:\n(\w+)|\b(red|blue|black|white|silver|gr[ae]y|burgundy)\b", text, re.I)
```

How it works: Two alternatives, each with its own group. re.search tries the field first at each position; m.group(1) or m.group(2) gives whichever one matched.

Watch out: Compare the hit count with 6.3: that difference is how many colors you rescued from free text.

Tested: 35 of 40 Hartford ads → Gray, red, blue

7. Location

Where is the car? The title line ends with Town, ST - craigslist, and sellers add their own clues.

7.1 ★★★ Town from the title line

```
re.search(r" - (.*?) , CT - craigslist", text)
```

How it works: .+? (lazy) grabs the town and stops at the first , CT.

Watch out: It misses towns in MA or NY. See 'Town AND state' below.

Tested: 39 of 40 Hartford ads → Plainville, Bolton, Southington

7.2 ★★★ The Craigslist area

```
re.search(r"CL\n(\w+)\n>", text)
```

How it works: The breadcrumb at the top: `CL`, then the area (`hartford`), then `>`. Useful once the whole class combines datasets!

Tested: 40 of 40 Hartford ads → `hartford`

7.3 ★★★ A list of towns

```
re.search(r"\b(hartford|new britain|bristol|manchester|enfield|vernon)\b", text, re.I)
```

How it works: Plain alternation. Easy to read, but only as good as your list.

Tested: 40 of 40 Hartford ads → `hartford`, `New Britain`, `Hartford`

7.4 ★★★ Town AND state

```
re.search(r" - ([^\n+), ([A-Z]{2}) - craigslist", text)
```

How it works: `[^\n+]` = anything except a comma or a line break. `[A-Z]{2}` = exactly two capital letters, which is a state code.

Tested: 40 of 40 Hartford ads → `('Plainville', 'CT')`, `('Longmeadow', 'MA')`, `('Bolton', 'CT')`

7.5 ★★★ Neighborhood in parentheses

```
re.search(r"^(((^\n+)\))$)", text, re.M)
```

How it works: `\(` and `\)` are **escaped** parentheses (real ones, not a group). In between, `[^\n+]` = anything except a closing paren. It finds `(Bolton)` and the seller's typo `(Plaiville)`.

Tested: 34 of 40 Hartford ads → `Plaiville`, `Bolton`, `Southington`

7.6 ★★★ Compass towns

```
re.search(r"\b(East|West|South|North) ([A-Z][a-z+])\b", text)
```

How it works: `[A-Z][a-z+]` = one capital letter, then lowercase letters, which is a proper name. It finds East Hartford, West Granby and South Windsor.

Watch out: Case-sensitive on purpose, so `south of the border` won't count.

Tested: 10 of 40 Hartford ads → `('East', 'Hartford')`, `('West', 'Granby')`, `('South', 'Windsor')`

7.7 ★★★ 'Located in ...'

```
re.search(r"located in ([A-Z][\w ]+?)(?:[.,\n]|$)", text, re.I)
```

How it works: Grab the words after `located in` up to the first period, comma or line break. `(?:[.,\n]|$)` is a stopping point we do not keep.

Tested: 3 of 40 Hartford ads → `Avon`, `South Windsor`, `CT`

7.8 ★★★ 'Location : ...' lines

```
re.search(r"^location\s*:\s*(.+)$", text, re.I | re.M)
```

How it works: `\s*` around the colon forgives `Location : Bristol` as well as `Location:Bristol`.

Tested: 1 of 40 Hartford ads → `Bristol/ Plainville / New Britain CT`

7.9 ★★★ Split a list of towns

```
re.split(r"\s*/\s*", "Bristol/ Plainville / New Britain CT")
```

How it works: `re.split` cuts a string wherever the pattern matches. Slashes with any spacing become a clean list of towns.

Result: `['Bristol', 'Plainville', 'New Britain CT']`

7.10 ★★★ In-state or out-of-state?

```
re.search(r", (?!CT\b)([A-Z]{2}) - craigslist", text)
```

How it works: Negative lookahead again: a state code that is NOT `CT`. It flags the cars you would have to drive to Massachusetts to see.

Tested: 1 of 40 Hartford ads → `MA`

8. What the seller says (flags for your model)

Turn free text into yes/no features: new parts, problems, owners, emissions. This is feature engineering with regex.

8.1 ★★★ New tires?

```
re.search(r"new tires", text, re.I)
```

How it works: The simplest possible feature: 1 if found, 0 if not. `bool(re.search(...))`.

Tested: 6 of 40 Hartford ads → `New Tires`, `new tires`, `New tires`

8.2 ★★★ New tires, brakes or battery

```
re.search(r"\bnew (tires|brakes|battery)\b", text, re.I)
```

How it works: One group, three options. `.group(1)` tells you which one was mentioned.

Tested: 12 of 40 Hartford ads → `BRAKES`, `BATTERY`, `brakes`

8.3 ★★★ Problem words

```
re.search(r"\b(rust|dents?|crack|damage|leaks?)\b", text, re.I)
```

How it works: `dents?` and `leaks?` make the `s` optional.

Watch out: It also matches `Rust Free` and `NO LEAKS`, which are good news! See 'Problems that are NOT denied' below.

Tested: 9 of 40 Hartford ads → `LEAKS`, `dents`, `Rust`

8.4 ★★★ One owner

```
re.search(r"\b(1|one|original) owners?\b", text, re.I)
```

How it works: Alternation for the three ways people say it.

Tested: 5 of 40 Hartford ads → `1`, `original`

8.5 ★★★ Everything that is 'new'

```
re.findall(r"\bnew (\w+)", text, re.I)
```

How it works: `findall` returns every word that follows `new`. Count them, or look at the most common ones with `collections.Counter`.

Watch out: `New Britain` is a town, not a new part. Every regex feature needs a quick look at what it caught.

Tested: 18 of 40 Hartford ads → `['Britain']`

8.6 ★★★ What it NEEDS

```
re.findall(r"\bneeds? (?!(nothing))(\w+(?: \w+)?)", text, re.I)
```

How it works: `needs?` matches `need` or `needs`. `(?!nothing)` skips the brag `Needs Nothing`. `(?: \w+)?` optionally grabs a second word: `new alternator`, `some TLC`.

Tested: 7 of 40 Hartford ads → `['Just Small']`

8.7 ★★★ Emissions done

```
re.search(r"emissions?\s+(?:done|ready|passed)|passed\s+(?:ct\s+)?emissions", text, re.I)
```

How it works: Two word orders joined by `|`: `emissions done` and `Passed CT Emissions`. `(?:ct\s+)?` = optional `CT`.

Tested: 11 of 40 Hartford ads → `Emissions passed`, `passed emissions`, `emissions done`

8.8 ★★★ Service records count

```
re.search(r"(\d+) service records", text, re.I)
```

How it works: A number in front of a phrase. `102 Service Records` becomes the feature `102`.

Tested: 3 of 40 Hartford ads → `102`, `60`, `29`

8.9 ★★★ Problems that are NOT denied

```
re.search(r"(?<!no )(?<!no fluid )\b(rust|dents?|crack|damage|leaks)\b(?! free)", text, re.I)
```

How it works: `(?<!no )` = negative **lookbehind**: skip it if `no` comes right before. `(?! free)` skips `rust free`. Negation is the hardest part of text mining, and part of why A07 hands this job to an LLM.

Tested: 5 of 40 Hartford ads → `dents`, `Crack`, `rust`

8.10 ★★★ SHOUTING lines

```
re.findall(r"^[A-Z0-9 ,.'!&/()-]{15,}$", text, re.M)
```

How it works: `{15,}` = 15 or more characters from the class (capitals, digits, punctuation, no lowercase). Count the ALL-CAPS lines: a writing-style feature that often points to the same repeat seller.

Tested: 11 of 40 Hartford ads → `'2008 JEEP WRANGLER'`, `'BLUE/TAN LEATHER INTE...`

9. Clean-up and privacy (re.sub)

`re.sub(pattern, replacement, text)` finds AND replaces. It is the tool for tidying text before a model or an LLM sees it, and for scrubbing personal information.

9.1 ★★★ Remove commas

```
re.sub(r",", "", "170,000")
```

How it works: The simplest `re.sub`: find `,`, replace it with nothing. `170,000` becomes `170000`.

Result: `170000`

9.2 ★★★ Collapse extra spaces

```
re.sub(r"[ \t]+", " ", "ratrod 327 v8")
```

How it works: `[ \t]+` = one or more spaces or tabs. Replace each run with a single space.

Result: `ratrod 327 v8`

9.3 ★★★ Fix a known typo

```
re.sub(r"\bhyunda\b", "hyundai", "2013 Hyunda Sonata", flags=re.I)
```

How it works: Whole-word replace. The `\b` matters: without it you would turn `hyundai` into `hyundaii`!

Result: `2013 hyundai Sonata`

9.4 ★★★ Fix a family of typos

```
re.sub(r"\binteriou?r\b", "interior", "TAN LEATHER INTERIOUR", flags=re.I)
```

How it works: `interiou?r` matches both spellings, and both become `interior`.

Result: TAN LEATHER interior

9.5 ★★★ Remove emojis and odd symbols

```
re.sub(r"[^\x00-\x7F]+", "", "Emissions Ready👍🚗")
```

How it works: `[^\x00-\x7F]` = any character that is NOT plain ASCII. The checkmarks and car emojis vanish.

Watch out: It also deletes accented letters (José) and curly quotes. Fine for a regex feature, but don't do this to text you show people.

Result: Emissions Ready

9.6 ★★★ Keep only the description

```
re.search(r"QR Code Link to This Post\n(.*)\npost id:", text, re.S)
```

How it works: `re.S` (DOTALL) lets `.` match line breaks too. The lazy `.*` stops at the FIRST `post id:`. What is left is just the seller's words: a much smaller, cleaner input for an LLM.

Tested: 40 of 40 Hartford ads → Selling my 2020 Honda HR-V Sport. Reliable, p..., Custom 1966 c10 tons of hand painted on it pi..., 2004 Pontiac Sunfire Bright Red, 140 HP at 56...

9.7 ★★★ Cut the header junk

```
re.sub(r"\A.*?QR Code Link to This Post\n", "", text, flags=re.S)
```

How it works: Same landmark, the other direction: delete everything from the start of the file (`\A`) up to the landmark.

Tested: changes 40 of 40 Hartford ads

9.8 ★★★ Redact phone numbers

```
re.sub(r"(?<!id: )\((?\\b\d{3}\\)?[-.\s]?\\d{3}[-.\s]?\\d{4}\\b", "[PHONE]", text)
```

How it works: Optional parentheses around the area code, optional separators (`[-.\s]?`), then 3 + 4 digits. Replace with `[PHONE]` before you share or store text.

Watch out: Without `(?<!id: )` it also "redacts" the 10-digit `post id` in every single ad. Test your cleaner, not just your extractor.

Tested: changes 0 of 40 Hartford ads

9.9 ★★★ Redact email addresses

```
re.sub(r"[\w.+-]+@[ \w-]+\.[\w.]+", "[EMAIL]", text)
```

How it works: Word characters, dots, plus signs, then `@`, then a domain with at least one dot. A classic, and a responsible habit.

Tested: changes 0 of 40 Hartford ads

9.10 ★★★ Catch sneaky phone numbers

```
re.search(r"\d*(?:zero|one|two|three|four|five|six|seven|eight|nine)\d*[-\s]?d{3}[-\s]?d{2}[-\s]?d{2}",
```

How it works: Some sellers write digits as words to dodge filters (like `86zero-...`). This pattern allows a number word mixed into the digits.

Watch out: There is always another trick. Regex catches the patterns you thought of, and that is exactly the case for trying an LLM in A07.

Tested: 1 of 40 Hartford ads (matches hidden: personal or identifying info)

10. Make and model (the boss level)

The field the extractor gets wrong. Build up to it here, then finish it in **Lab 3, Way 2**.

10.1 ★★★ One brand

```
re.search(r"honda", text, re.I)
```

How it works: Plain text plus `re.I`. It works, but you would need one pattern per brand.

Tested: 12 of 40 Hartford ads → `honda`, `Honda`

10.2 ★★★ A list of brands

```
re.search(r"\b(honda|toyota|ford|chevy|chevrolet|bmw|jeep|acura|hyundai|nissan)\b", text, re.I)
```

How it works: Alternation with word boundaries. Only as good as your list: it never finds a Jaguar.

Tested: 34 of 40 Hartford ads → `honda`, `chevy`, `jeep`

10.3 ★★★ Both spellings of Chevy

```
re.search(r"\bchev(y|rolet)\b", text, re.I)
```

How it works: Put the shared start outside the group and the different endings inside: `chev` + (`y` OR `rolet`).

Tested: 4 of 40 Hartford ads → `chevy`, `chevrolet`, `Chevrolet`

10.4 ★★★ Hyphen or no hyphen

```
re.search(r"\b(cr-?v|hr-?v|rav-?4|f-?150)\b", text, re.I)
```

How it works: `-?` = an optional hyphen. `CRV`, `CR-V` and `cr-v` all match.

Tested: 6 of 40 Hartford ads → `hr-v`, `f-150`, `cr-v`

10.5 ★★★ First word after the title year

```
re.search(r"\A\d{4} (\w+)", text)
```

How it works: Start of file, the year, a space, then capture one word. That is usually the make.

Watch out: The title is typed by the seller: `FOR FOCUS`, `Hyunda`. Garbage in, garbage out.

Tested: 40 of 40 Hartford ads → `honda`, `chevy`, `Pontiac`

10.6 ★★★ Make AND model from the title

```
re.search(r"\A\d{4} (\S+) (\S+)", text)
```

How it works: `\S` (capital S) = anything that is NOT whitespace. Unlike `\w`, it keeps the hyphen in `hr-v`.

Tested: 40 of 40 Hartford ads → `('honda', 'hr-v')`, `('chevy', 'c10')`, `('Pontiac', 'Sunfire')`

10.7 ★★★ The whole title before the boilerplate

```
re.search(r"\A(.+?) for sale by owner", text)
```

How it works: Everything from the start of the file up to `for sale by owner`. Good for eyeballing and for feeding an LLM.

Tested: 40 of 40 Hartford ads → `2020 honda hr-v sport`, `1966 chevy c10 truck`, `2004 Pontiac Sunfire`

10.8 ★★★ The model after a known make

```
re.search(r"\b(?:honda|toyota|ford|jeep|acura)\s+(\w-+)", text, re.I)
```

How it works: Match the make without capturing it, then capture the next word. `[\w-]+` = word characters or hyphens.

Tested: 25 of 40 Hartford ads → `hr-v`, `wrangler`, `f-150`

10.9 ★★★ Why the current extractor fails

```
re.search(r"\b([A-Z][a-z]+)\s+([A-Z][A-Za-z0-9]+)", text)
```

How it works: This is the OLD `MAKE_MODEL_RE` in your pipeline: any capitalized word followed by another capitalized word. The first place that happens on the page is often `Contact Information`. Run it and see!

Watch out: The lesson: a regex that works on one example can fail on the real page. Always test on many ads.

Tested: 40 of 40 Hartford ads → ('Contact', 'Information'), ('Pontiac', 'Sunfire'), ('New', 'Britain')

10.10 ★★★ Lab 3 Way 2: the details-block version (you finish it)

```
re.search(r"^(?:19|20)\d{2}\n(____)\s+(____)", text, re.M)
```

How it works: Combine idea 2.5 (a line that is only a year) with a line break and two capture groups (make, model). The year line is written as `^(?:19|20)\d{2}\n`. The two groups are your job in Lab 3.

Tested: Dr. Wanik's answer matches **40 of 40** Hartford ads. Can yours?

Where these ideas go next

- **Lab 3, Way 2:** finish idea 10.10 and put it in `BETTER_MAKE_MODEL_RE`.
- **A06:** ship your best make/model regex to your pipeline (one line, on a branch, then a pull request).
- **A07:** Gemini reads the same ads. Where does it beat your regex (negation, typos, sneaky phone numbers), and where does your regex win (speed, cost, exact fields)?
- **Midterm:** sections 5, 6 and 8 are full of new features for your price model: photos, one owner, new tires, service records, color.